DATA INTERPRETATION

- Data Sufficiency
- Data interpretation-Advanced
- Interpretation tables, pie charts & bar chats

DATA INTERPRETATION

Data interpretation is part of daily life for most people. Interpretation is the process of making sense of numerical data that has been collected, analyzed, and presented.

As we know, the Data can be represented in four forms
1. Numerical: Data in numerical form
2. Table: Data in table form
3. Mixed: Data in mixed form
4. Graphical form like Bar, Pie, Line etc.,

KEY POINTS

- Be clear on what is given and what is asked.
- See how much effort is required to get the required data from the given data.
- If a question demand unreasonable efforts, LEAVE those questions and come back if possible.
- Focus only on useful data
- Never lose your heart if you see lot of data. Until you are able to comprehend the given data, you are good to go.
- Do ONLY what is required. Don't try to keep data ready before solving!
- Approximate wherever possible.
- If only relative values are asked, don't calculate the actual values.

IMPORTANT

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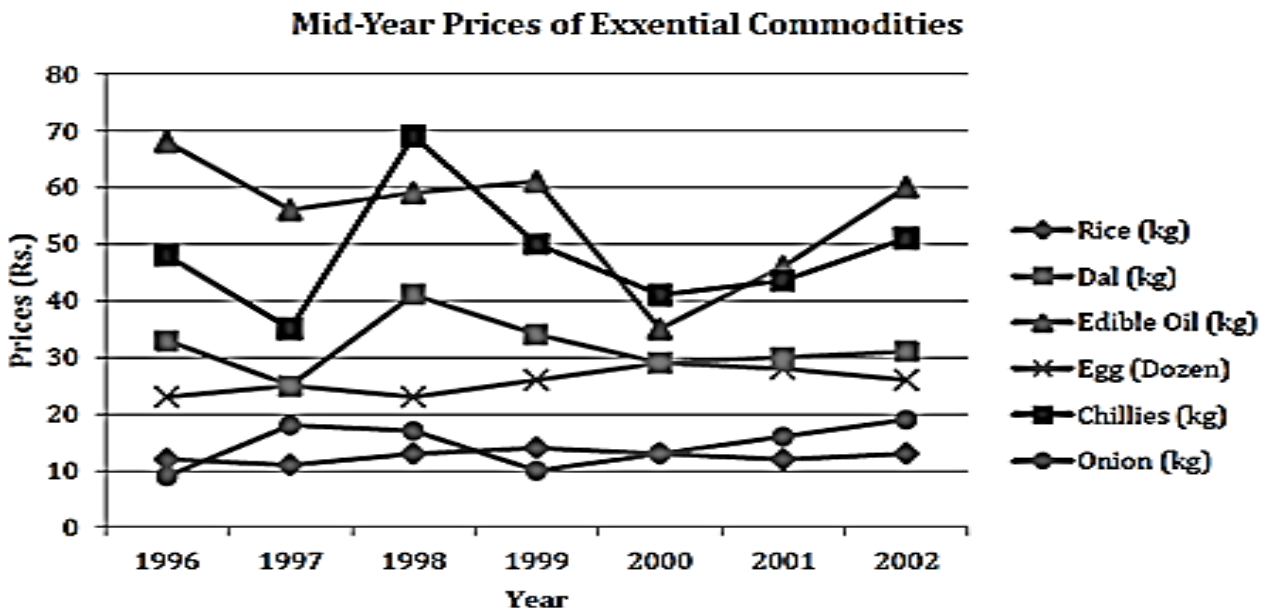
EXAMPLE PROBLEM



Directions (Q. 1 to 5): Study the table below to answer these questions.

1. During 1996-2002, the number of commodities that exhibited a net overall increase and a net overall decrease, respectively, were

A. 3 and 3
B. 2 and 4
C. 4 and 2
D. 5 and 1

2. The number of commodities that experienced a price decline for two or more consecutive years is

A. 2 B. 3 C. 4 D. 5

3. For which commodities did a price increase immediately follow a price decline only once in this period?

A. Rice, Edible oil, and Dal
B. Egg and Dal

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C. Onion only
D. Egg and Onion

Explanatory Answer:

What is given?

Data is represented using a Qualitative method (Line Graph). Given data are

Price of commodities from year 1996 to 2002
Trend of commodity price during the tenure 1996 – 2006

What is asked?

- Net overall increase/decrease of commodities during the period

- Trend in price fluctuations

Effort required:

Question 2 & 3 are purely qualitative in nature. Visual inspection can yield the answer. Both are very much doable and can fetch us some easy marks.

Question 1, we have to find the net increase/decrease of price for the period. We can either approach it using conventional quantitative methods or can use a qualitative approach. If the qualitative approach crossed your mind by now, it's all yours... else don't waste time in this one unless you don't have an easier question left to do.

As a matter of fact, you don't have to touch your pen to solve all the three questions.

Question #2

By visual inspection, answer is five. Except for edible oil all other commodity price decline for two or more consecutive years during the tenure. We need to find the pattern of 2 continuous dips in the graph for the commodity during the period.

Question #3

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This question needs us to find the commodities which has a price decline followed by a price increase and only once. We need to find a particular pattern in the plot which is a dip and rise coming together and only once. A Dip-Rise together makes a V shaped plot. Hence we are asked to identify those commodities whose plot has one and only one V in it.

Before start solving, we will see another important aspect here. Options are there for helping us. Not just the right one, but all options! Build the habit to bring down the calculations to the minimum and do only what is really required. Less calculations means lesser time required and lesser chance for committing mistakes.

Carefully observe the options to see whether there is any chance to simplify our efforts. Here Option 3 needs us to find the pattern only for one commodity (Onion). The trick here is that if Onion satisfies the condition, then we can eliminate options 1 and 2 as Onion is not there in both of them. Otherwise if Onion does not satisfy the condition then we can remove option 3 and 4 as Onion is included in them.

Onion has one V and only one V in its plot. Option 1 and 2 out from our attention.

Now to mark between option 3 and 4 we need to take the plot of Egg. We have one and only one V in egg plot too... Hence option 4 is the right one.

Question #1

Many mistakes we commit during our aptitude exam is not mathematical, it is psychological. By using best method
or not, it is relatively easy to solve the 2nd and 3rd questions in this set. Once we get those two questions into our basket it is natural to urge to 'Conquer' the whole set by solving the first question. First question is also easy if we got the correct approach. Otherwise we are in loss in terms of time.

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We are asked to find the net increase/decrease in the overall price in the period. There is no optimization available as we need to find the status of all the commodities to get the answer.

We can go with the quantitative method, like Rice price fluctuation during the period:
10 -> 14 -> 13 -> 10 -> 11 -> 12 -> 14
Net price difference = +4 – 1 – 3 + 1 + 2 + 2 = +4.
(Hence net is positive)

PRACTICE PROBLEMS (EXPLANATORY ANSWERS AT THE END)

TABLE BASED:

Directions for questions 1 to 5: Refer to the data in the TABLE given below.

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Given below is the performance of scores in "University Exams" of students of 5 schools viz. A, B, C, D and E, in 3 subjects. The numbers indicate how many students from each of the schools have passed in a particular subject. The last column gives details of the total number of students who took the examination from that school.

School	Math	Chemistry	Physics	Total number of students
A	450	520	400	550
B	230	400	350	450
C	450	475	500	600
D	340	360	420	500

1. How many students from school A have passed both in Math and Physics?

A. 300
B. 450
C. 520
D. Data Insufficient

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2. What is the percentage of students in school C who have passed in Physics?

A. 66.66%
B. 80 %
C. 83.33%
D.86.66%

3. The school with the highest pass percentage in Math is?

A. A
B. B
C. C
D. E

4. Which subject has the highest pass percentage when all schools are considered?

A. Math
B. Chemistry
C. Physics
D. Indeterminate

5. What is the approximate percentage of number of students in School A who have passed in Physics compared to the total number of students who have passed in Physics from the 5 schools put together?

A. 18.7%
B. 22.5%
C. 12.5%
D. 16.66%

LINE GRAPH BASED:

Directions for questions 6 to 10: Refer to the data in the LINE GRAPH given below.

The graph indicates the value of a car purchased today (year 0) over the years. Assume that the price of a new car remains the same for eternity. Ignore time value of money for all calculations.

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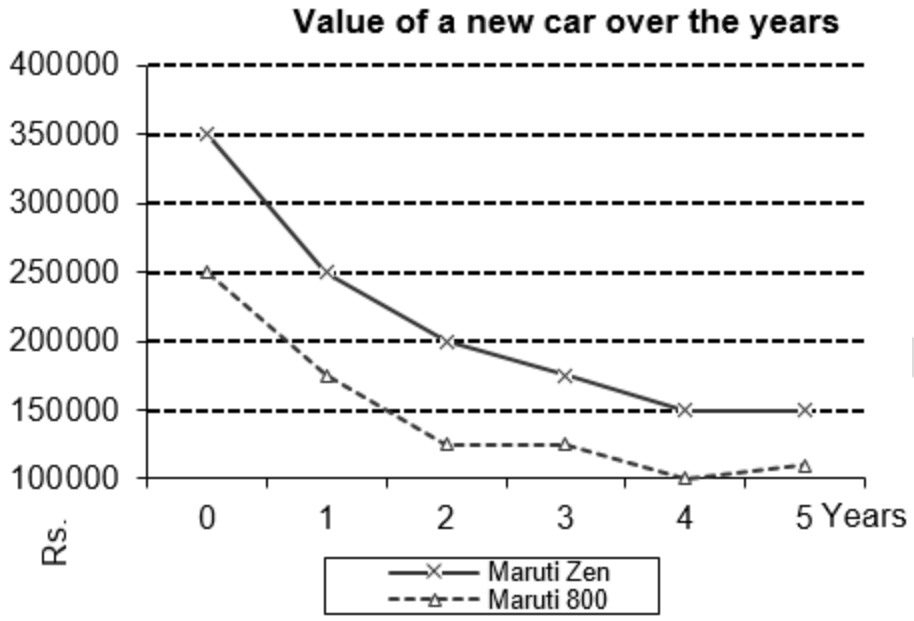

6. How much more expensive is a Maruti Zen over a Maruti 800 at the beginning of year 0?

A. 30%
B. 35%
C. 40%
D. Indeterminate

7. What is the average annual depreciation for a Maruti Zen over the period (5 years)?

A. 9.5%
B. 11.5%
C. 13.5%
D. Indeterminate

8. In which year is the price differential between the 2 cars the least?

A. Year 3
B. Year 5
C. Year 4
D. Indeterminate

9. A good investment is defined as lesser depreciation over a specified period of time. Which car serves as a better investment option?

A. Maruti 800

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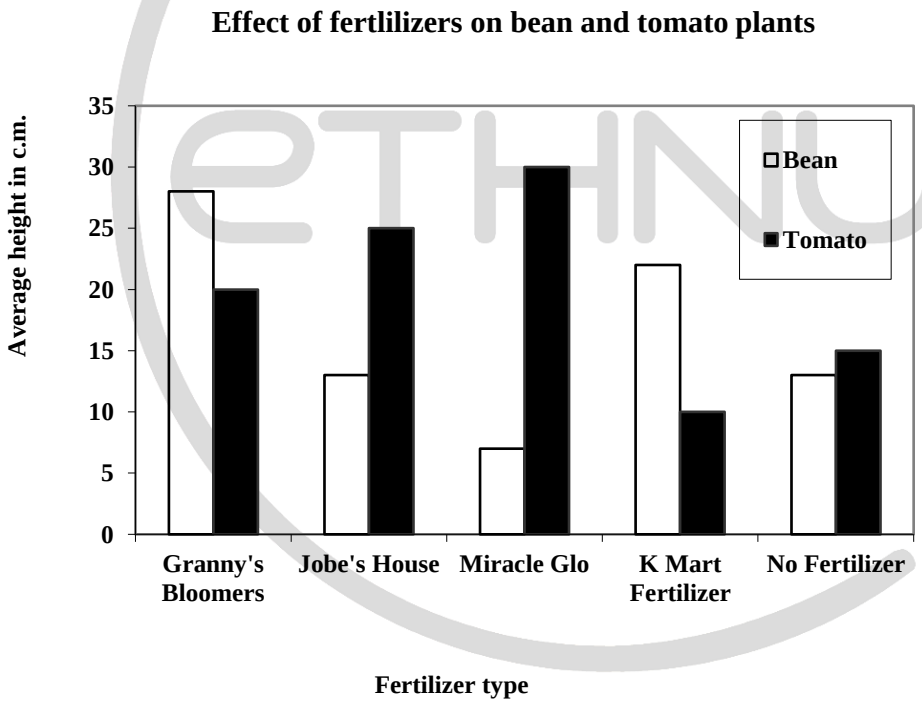
B. Maruti Zen
C. Both Equally Good
D. Indeterminate

10. If I buy a Maruti Zen, sell it at year 1 and buy a new Maruti 800, how much money will I be left with?
A. Rs.60000
B. Rs.65000
C. Zero
D. Indeterminate

BAR GRAPH BASED:

Directions for questions 11 to 15: Refer to the data in the BAR GRAPH given below.



Note:

- Height is an indicator of effectiveness of fertilizer.
- Equal amounts of fertilizer are required for each application.
- Price of all fertilizers is Rs.30/- per kg
- Height of plants is not necessary proportional to the ratio of fertilizers mixed.

11. If a farmer growing bean and tomato plants in equal numbers could buy only one brand of

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fertilizer, then which brand would he opt for maximum effectiveness?

A. Granny's Bloomers
B. Jobe's House
C. Miracle Glo
D. Indeterminate

12. How much superior is Miracle Glo when compared to Jobe's House for tomato plants?

A. 5%
B. 16.67%
C. 20%
D. Indeterminate

13. If I am a tomato farmer and I decide to use Miracle Glo, how much will I have to spend for 10 tomato plants?

A. Rs.870
B. Rs.900
C. Rs.930
D. Indeterminate

14. If a bean farmer uses a mixture that contains equal quantity of all fertilizers, then what will be the average height of his bean plants?

A. 17 cm
B. 18 cm
C. 21 cm
D. Indeterminate

15. If a tomato farmer currently does not use fertilizer, then how much better off is he if he decides to use Miracle Glo?

A. 50%
B. 100%
C. 200%
D. Indeterminate

PIE CHART BASED:

Directions for questions 16 to 20: Refer to the data in the PIE CHART given below.

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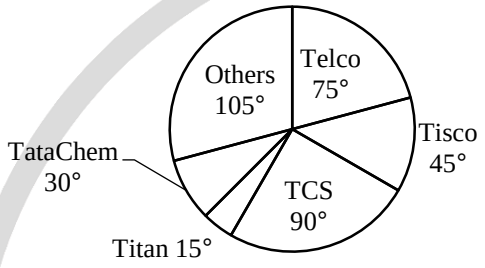
Contribution of Various Tata Group companies to the profits of the holding company - Tata Sons (Total = 2500 crores)



16. What was the contribution of Tisco to the total profits of Tata Sons?

A. Rs. 312.5 crores
B. Rs. 112.5 crores
C. Rs. 125 crores
D. Rs. 375 crores

17. Contribution of TCS towards the total profit exceeded contribution of Tata Chem by how much percentage?

A. 100 %
B. 200 %
C. 300%
D. 600%

18. If Titan was to be represented under the head "Others", then what would be the percentage increase in the area under the head "Others"?

A. 14.25%
B. 16%
C. 18%
D. 11.5%

19. I in the next year, all companies showed no increase in contribution of profits and only TCS increased its contribution by 50% then what

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would be the measurement of the arc subtended by the TCS portion to the center of the pie?

A. 90°
B. 145°
C. 120°
D. 75°

20. If contribution of profits by TCS increased by 50%, then by what percentage should Telco increase its contribution to profits to retain the same angle, given that only these two companies showed an increase in profit contribution to Tata Sons?

A. 50%
B. 75%
C. 17.5%
D. 15.8%

CASELET BASED:

Directions to questions 21 to 25: Refer to the data in the CASELET given below.

With the entry of XYZ in 1999, USA saw the sales of all other major English dailies falling. XYZ knocked off 5% of the market shares of PQR and ABC each, while LMN lost 1/5th of its readers to XYZ. In the next year, XYZ had captured a quarter of the market, while PQR remained unchanged. While LMN further lost 1/4th of its readers to occupy 30% of the market, ABC sold more newspapers and got back its market share of 1998. The number of readers in this time period did not increase, and assume that there were no other English dailies. In 1998, PQR sold 1/4th the newspapers sold by ABC.

21. What was the average market share of LMN for the 3 years?

A. 25%
B. 40%
C. 33.33%
D. 30%

22. Sales of PQR in 1998 were _____ sales of LMN in the same year.

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A. 1/3
B. 1/2
C. 1/4
D. 1/5

23. What was the contribution of XYZ in the total sales of English newspapers in USA for the years 1999 and 2000?

A. 22.5%
B. 45%
C. 25%
D. 21%

24. For how many years in the 3-year period, were sales of ABC equal or higher than sales of LMN?

A. Never
B. One
C. Two
D. Three

25. What was the approx. average market share of ABC for the three years?

A. 39%
B. 40%
C. None of these
D. Indeterminate

SOLUTION WITH EXPLNATORY ANSWER

Solutions to questions 1 to 5:

1. As no information is provided about English, the data is insufficient.
Hence, the correct answer is **options (4).**

2. 500/600 x 100 = 83.33%.
Hence, the correct answer is **option (3).**

3. It is easy to see that the pass percentages in schools B and C are the highest.
Pass% in Math in school A = $\frac{450}{550}$ x 100; in school E = $\frac{410}{500}$ x 100
To find out which of $\frac{450}{550}$ and $\frac{410}{500}$ i.e., $\frac{9}{11}$ and $\frac{41}{50}$ is the highest.

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Doing cross multiplication, 9 × 50 = 450, and 11 × 41 = 451.
As 451 > 450, $\frac{41}{50} > \frac{9}{11}$. Therefore school E has the highest %.
Hence, the correct answer is **option (4).**

4. To find this, we just have to find which sum among Math, Chemistry or Physics is the highest.
It is easy to see that Mathis not the answer.
Total number of students passed in Chemistry = 2205
Total number of students passed in Physics = 2140
Hence, the correct answer is **option (2).**

5. The required % = $\frac{400}{2140}$ × 100 ≈18.7%.
Hence, the correct answer is **option (1).**

Alternate solution:
$\frac{400}{2000}$ x 100 = 20%; $\frac{400}{2400}$ x 100 = 16.67%.
The answer should lie between 16.67% and 20%.
As 18.7 is the only such option, that is the answer.

Solutions for questions 6 to 10:

6. $\frac{35-25}{25}$ x 100 = $\frac{2}{5}$ = 40%
Hence, the correct answer is **option (3).**

7. Average annual depreciation = ($\frac{35-15}{35}$ x 100)/5 ≈ 11.5%
Hence, the correct answer is **option (2).**

8. From the graph, we can see that in year 5, the difference between the prices is the least.
Hence, the correct answer is **option (2).**

9. As the period of investment is not mentioned, the question can't be answered.
Hence, the correct answer is **option (4).**

10. Maruti Zen can be sold for Rs. 2,50,000 after 1 year. A new Maruti 800 can be bought for Rs. 2,50,000.

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So, no money would be left after buying a Maruti 800 from the money obtained after selling 1 year old Zen.
Hence, the correct answer is **option (3).**

Solutions for questions 11 to 15:

11. From the figure, on an average, only Granny Bloomers is good for both the plants.
Hence, the correct answer is **option (1).**

12. As the superiority is directly related to the height, Miracle Glo is superior to Jobe's House by $\frac{30-25}{25}$ x 100 = 20%.
Hence, the correct answer is **option (3).**

13. Since the cost of tomato plant is not given, the question cannot be answered.
Hence, the correct answer is **option (4).**

14. As it is mentioned that height of a plant is not necessarily proportional to the ratio of fertilizers mixed, the question cannot be answered.
Hence, the correct answer is **option (4).**

15. The farmer is better off by $\frac{30-15}{15}$ x 100 = 100%.
Hence, the correct answer is **option (2).**

Solutions to questions 16 to 20:

16. Contribution of Tisco (in crore Rs.)= $\frac{45}{360}$ x 2500 = 312.5.
Hence, the correct answer is **option (1).**

17. In any question where it is required to find the percentage finally, the actual values of the elements involved in the question does not matter.
So, for convenience one can assume the total value of the profits as 360 units. In that case, the profit of TCS and Tata Chemicals would be 90 units and 30 units respectively.
The required % = $\frac{90-30}{30}$ x 100 = 200%.

18. Similar to the previous, the actual profits 'others' does not matter.

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% change = $\frac{final-initial}{initial}$ x 100 = $\frac{120-105}{105}$ x 100
= $\frac{15}{105}$ x 100 ≈ 14.25%.
Hence, the correct answer is **option (1).**

19. Let us assume that the total profits as 360 units.
Then the profits of TCS would be 90 units.
When increased by 50%, the profits of TCS would become 90 + 45 = 135 units.
As the profits of no other company have increased, the profits of Tata Sons would become 360 + 45 = 405 units.
The share of TCS after increase in profits = $\frac{135}{405}$ x 360 = 120°.
Hence, the correct answer is **option (3).**

20. If TCS increases its profits by 45 units and Telco by 'x' units the profits of Tata sons would increase by 360 + 45 + x = (405 +x) units.
As the percentage contribution of Telco does not change,
$\frac{75+x}{405+x}$ × 360 = 75 ⇔ $\frac{75+x}{405+x}$ = $\frac{5}{24}$ ⇔ 1800 + 24x = 2025 + 5x
⇔ 19x = 225 ⇔x = 225/19
Let the required % be 'y'.
y%(75) = 225/19 ⇔ y = 300/19 ≈ 15.8%
Hence, the correct answer is **option (4).**

Solutions for questions 21 to 25:

Assuming that the total sales of all the newspaper put together as 100 per year, the number of newspapers sold by different dailies in the 3 years is as shown below

	XYZ	PQR	ABC	LMN
1998	-	10	40	50
1999	20	5	35	40
2000	25	5	40	30

Let the number of dailies sold by LMN in 1999 be 'x'. It lost 1/4th of the market share and became 30.
Hence, x – ¼(x) = 30 ⇔ ¾ (x) = 30 ⇔ x= 40.
Similarly, share of LMN in 1998 = y – 1/5 (y) = 40 ⇔ y = 50.

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PQR sold 1/4th of that of ABC. Also dailies sold by PQR and ABC = 50. From this information, no. of dailies sold by PQR and ABC can be found to be 10 and 40 respectively.
As XYZ knocked off 5% of the market share of PQR and ABC, their shares in 1999 wound become 5 and 35 respectively.
As ABC remains its market share in 2000, it becomes 40.

21. Average market share of LMN for the 3 years = (50 + 40 + 30)/3
= 40.
Hence, the correct answer is **option (2)**.

22. 10 = n × 50 ⇔ n = 1/5.
Hence, the correct answer is **option (4)**.

23. The % contribution of XYZ in 1999 and 2000
$= \frac{20+25}{100+100} \times 100$
$= \frac{45}{200} \times 100 = 22.5\%$.
Hence, the correct answer is **option (1).**

24. The sales of ABC exceed the sales of LMN only in year 2000.
Hence, the correct answer is **option (2).**

25. The average share of ABC = (40 + 35 + 40)/3 ≈ 38.33%.

Though the answer is very close to 39%, it is not equal to 39%; and, since there is 'None of these' option, we should go for it.
Hence, the correct answer is **option (3).**

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